PAPER • **OPEN ACCESS**

Vehicle Real-time Condition Monitoring Based on CAN Bus

Vehicle Real-time Condition Monitoring Based on CAN Bus

Yuehao Wang[1*], Jian Xu[1]

[1]Tianjin University of Technology and Education School of Electronic Engineering, Tianjin, China

*Corresponding author: wangyuehao0825@163.com

Abstract: Because the automobile in the driving process there will be some hidden dangers, once the hidden dangers occur is likely to have a safety accident, then the driver's safety will be endangered. In order to avoid this situation, this paper introduces multi-sensor communication under the CAN bus, using the CAN interface of STM32F103 MCU to transmit data with multi-sensor in real time. The high performance and reliability of CAN have been recognized, and it has been widely used in the field of automobile communication. It provides powerful technical support for the real-time and reliable data exchange of distributed control systems among nodes. Displaying the final result on the display when communicating with multiple sensors allows the driver to monitor the status of the car in real time and process it in the first place, providing a safety guide for driving.

1. Introduction

Because of the continuous development of automotive electronic technology, the automobile has become an important part of our society, the safety of the driver is still a problem we need to solve, to be able to monitor the driving state of the car in real time, once there is a problem can be timely feedback to the driver, to ensure that they can drive safely[1]. Because the state information of some cars is difficult to be measured directly, the previous research usually used a single sensor to collect data through a single bus, and then fused the information, so the collected information is usually not real-time, and cannot guarantee the driver's safety well. At present, there is more and more domestic research on CAN bus in automobile control. The application of CAN bus technology plays an important role in the development of automotive electronic technology. In addition, multi-sensor communication is widely used in the automotive field, and it CAN collect data and information about sensors in many aspects, so in this paper, we use the common sensors to transmit the multi-sensor data in order of priority in CAN bus, transmit the information to VCU, and then display it on the monitor, this can not only ensure that the transmission of information will not be chaotic, can receive effective information, but also real-time monitoring of vehicle status information to ensure the safety of drivers[2][3].

2. Principle of real-time monitoring

Each sensor is connected to the STM32 microcontroller, and then the data is transmitted to VCU through the differential CAN_Hvoltage between CAN_H and CAN_L, two signal lines of the CAN bus, and the data value is displayed through the display. Its characteristic is that you can implement multiple sensors transmitted together, and CAN bus arbitration is according to the priority order of sensor corresponds to an orderly transmission. And CAN bus arbitration according to the sensor's corresponding priority order for orderly transmission, to ensure the real-time and effective transmission, and finally through the display will display the value, once an exception occurs, a warning sign is displayed, the driver will detect the vehicle to stop in time, and keep it safe[4][5].

The sensors used in this paper are DS18B20, MMA7361 and A3144[6], which are the most common temperature sensors, accelerometers and speed sensors. To ensure that the system to the engine temperature, angle and speed, such as a reasonable judgment of the numerical value, its safety threshold settings as shown in Table 1.

Table 1 Sensor security thresholds

sensor	temperature	acceleration	speed
Safety threshold	20-50 degrees	0-20 degrees	2000-6000 RPM

If the collected data is within the normal threshold, the system will directly display the data through the display screen. If the collected data is beyond the normal threshold, the system will judge the data as abnormal data, and abnormal symbols and abnormal data need to be displayed on the display screen to inform the driver. The transmission process is shown in Figure 1.

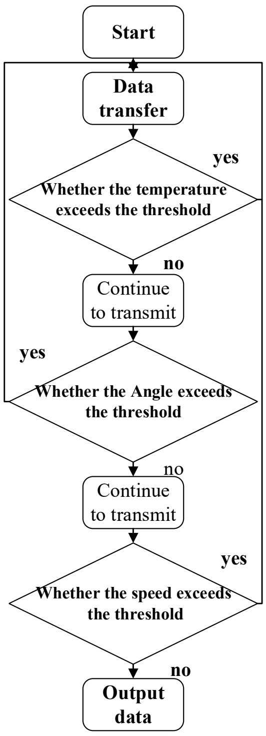

Figure 1 Data transfer process

The transmitted data is transferred to the display through the CAN bus in order of priority. According to the MAC mechanism of CAN and the arbitration rules, if the high-priority message is always transmitting, then the low-priority message CAN not get the arbitration, but tries to send the message again all the time. Only when the high-priority message no longer transmits data, can the low-priority message obtain the transmission authority, thus sending the message[7].

3. Sensor selection and design

3.1. Temperature sensor

DS18B20 is a commonly used digital temperature sensor. It is characterized by the use of a single pin that can complete the master and slave two-way communication. Its physical connection is a simple, flexible design, and it has a strong ability to adapt to the site. After the measurement, it can be directly displayed as a digital temperature signal and then sent to the single-chip microcomputer serial.

The temperature sensor can be directly connected with the STM32 single-chip microcomputer and connected with the PC0 interface of the single-chip microcomputer to realize the temperature collection directly by the single-chip microcomputer. The GND port is grounded, the UDD port is connected to a 5V power supply, and the DQ port is connected to the I/O port of the MCU. It communicates with the microcontroller through the DQ pin, and the initial configuration of DQ is required. Figure 2 below shows the hardware connection between DS18B20 and STM32F103.

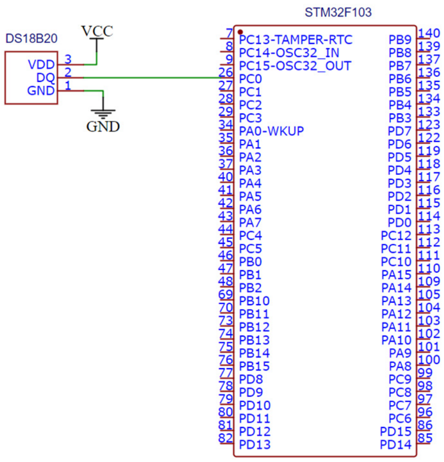

Figure 2 Hardware connection between DS18B20 and STM32F103

3.2. Acceleration sensor

The essence of the acceleration sensor is to cause the deformation of the sensitive parts inside the sensor through the force, and the corresponding acceleration signal can be obtained by measuring the deformation and converting it into voltage output by the relevant circuit. MMA7361 chip is used in this paper, which can detect the attitude or direction of motion of the object. It is characterized by long battery life, stable and reliable signal anti-interference ability, and high sensitivity. The pin function table corresponding to the chip during signal collection is shown in Table 2:

Table 2 Chip pin function list

NC	dangling
VSS	grounding
VDD	Connect the power supply
XOUT	X axis is used to detect the simulated quantity
YOUT	Y axis is used to detect the simulated quantity
ZOUT	Z axis is used to detect the simulated quantity
SLEEP	The default value is 1, so that the sensor can work, and can collect analog signals, and then through A/D turn and use the following formula to calculate the measured Angle value
SELF/TEXT	Whether to enable the self-check mode
G-SELECT	1: The range is 6g and the sensitivity is 206mV/g
	0: the range is 1.5g and the sensitivity is 800mV/g
0G-DELECT	1: free falling, can be connected to an alarm trigger signal
	0: indicates normal conditions

MMA7361 can only be measured by gravity. Its acceleration is measured by gravity, using the relationship between the angle and voltage, which can be measured relative to a plane tilt angle. The sensor is designed to connect with the MCU to collect the Angle information, as shown in Figure 3.

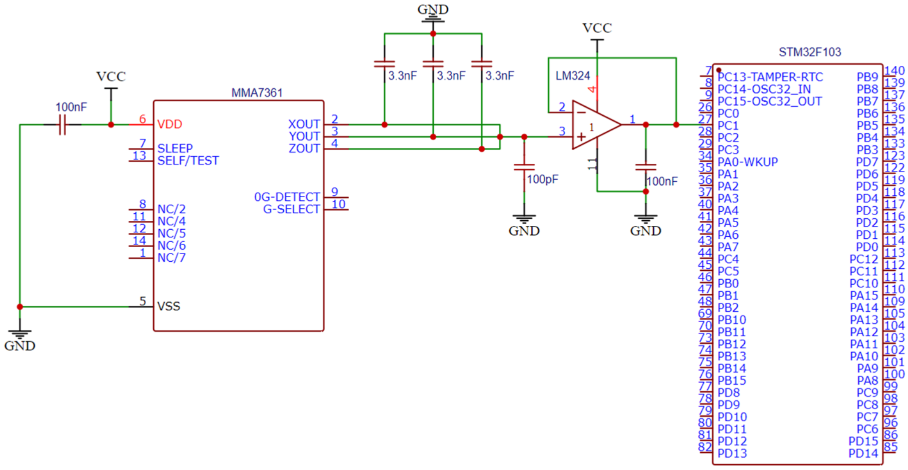

Figure 3 Hardware connections between the MMA7361 and STM32F103

For the collected signals, because of the acceleration sensor, there may be some noise and interference, so to carry on the processing, and output of X, Y, and Z pin for analog signals, so it must pass through the AD module for digital-to-analog conversion. A signal conditioning circuit is designed in the figure, which uses LM324 to improve the load capacity. Finally, it is input into the PC1 interface of the single-chip microcomputer.

3.3. Hall sensor

Hall sensor A3144 is used to measure the speed. When the motor rotates, the motor drives the Hall effect sensor, and a corresponding frequency pulse signal is generated. The pulse signal of each cycle reaches the count of the pulse through the Counter Register of the single-chip microcomputer; thus, the speed of rotation can be measured. VCC is connected to a 5V power supply, GND is grounded, and OUTPUT is the signal output end. Figure 4 shows the hardware connection diagram of A3144 and STM32F103.

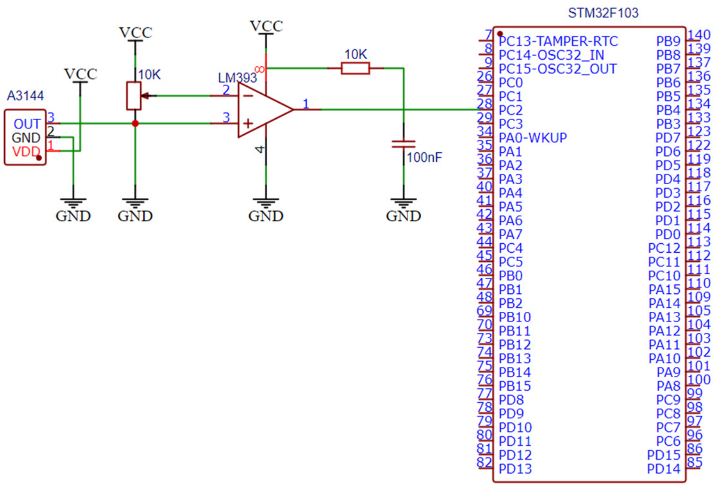

Figure 4 Hardware connections between the A3144 and STM32F103

Hall sensor also designed a signal conditioning circuit, the signal using voltage comparator LM393 and potentiometer shaping, 10K resistance is connected to the pull-up resistance, and then input it to the PC2 interface of MCU. The collected speed signal is transmitted to the single-chip microcomputer and then transmitted to the VCU via the CAN bus. Finally, the speed is displayed on display.

4. CAN communications with sensors

Since STM32 is widely used, and there are more I/O ports available for easy connection with other devices. It is necessary to connect a CAN transceiver to the CAN interface of STM32 for data transmission. TJA1050 is used in the CAN transceiver to convert the logic level signal transmitted by the CAN controller into a differential level signal of the CAN bus. Since it supports CAN2.0A and CAN2.0B protocols, it only needs to connect a CAN transceiver, TJA1050, to carry out data communication. Figure 5 below shows the designed hardware circuit.

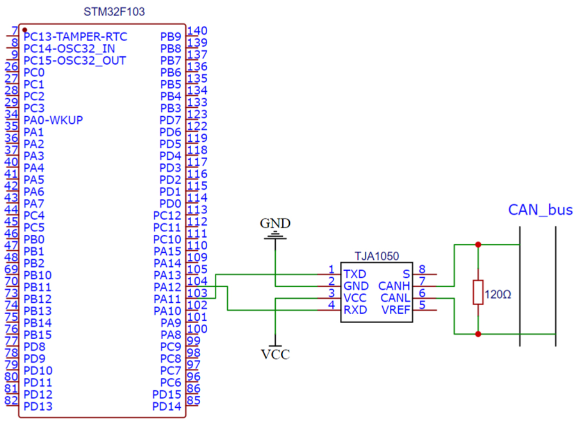

Figure 5 Connection between the STM32 and CAN transceiver

It can be seen from the figure that the CAN sending pin of STM32F103 is PA12(CAN_TX) and the receiving pin is PA11(CAN_RX). The serial output TXD and input RXD corresponding to the CAN transceiver are connected. The GND and RS pins of the transceiver are grounded, the VCC is connected to a 5V power supply, and the CAN and CAN are connected to the CAN bus. As the SAEJ1939 protocol is a protocol based on a CAN bus with a baud rate of up to 250 Kbps, it is a communication network protocol with a high transmission rate. Therefore, a reasonable communication protocol is formulated according to the SAEJ1939 protocol. First of all, the sensor should be address coded to formulate the communication protocol. Set the address of temperature sensor DS18B20 to 0X0A, acceleration sensor MMA7361 to 0X0B, and speed sensor A3144 to 0X0C. When the CAN bus receives the sensor information at the same time, it prioritizes them according to their address codes, prescribes that the CAN bus preferentially recognizes the temperature sensor code 0X0A, and then recognizes the accelerometer code 0X0B. Finally, the coding of the speed sensor is 0X0C. To ensure data accuracy, each node of the CAN bus should check data before sending messages, and only receive corresponding messages through network protocol when receiving messages. So they will not be the chaotic transmission, and can be in accordance with the priority of orderly transmission, to ensure real-time and effective information, more to ensure the safety of the driver.

The display is the last key link, needed in VCU outside a display, used for real-time display of sensor values. When all parameters are within the set threshold range, and the corresponding values are displayed, the car is safe and can be driven normally. Once the value is not within the threshold range, its data will be marked in red, and in the display interface will appear exclamation mark to prompt you about data problems. The driver will find the car problems in time, at this time to stop driving, to further solve, to maximize the safety of the driver. Figures 6 and 7 below show the values displayed by the display under normal conditions and the values displayed by the display when there is an abnormal value.

Figure 6 Normal display

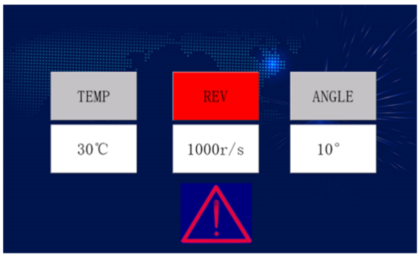

Figure 7 Abnormal display

5. Conclusions

This paper introduces sensor communication based on the CAN bus, and CAN is widely used in today's communication field. It is difficult to meet the requirements of driving safety when a single type of sensor detects information and makes decisions on the vehicle, and more and more sensors are installed now, so it is necessary to test their information reasonably. This paper is about multi-sensor information processing, each sensor information through the CAN bus transmission on display. Because there are more traffic accidents now, to protect the safety of drivers to a greater extent, we use multi-sensor communication to help drivers pay attention to the state of the car promptly, once there is a problem can be found and handled promptly, so not only to provide driving guidance, but also to ensure the safety of the driver's driving.

References

[1] Zhang Xin. Automotive safety system design and application of multi-sensor support [J]. Journal of Electronic and Information Technology, 2020, 4 (4): 9-10 + 19. DOI: 10.19772 / j.carol carroll nki. 2096-4455. 2020.4.004.
[2] WU Haichao. Design and Research of Automobile Fire Detection and Alarm System based on Multi-sensor Information Fusion [J]. Internal Combustion Engine and Accessories, 2021 (21): 212-213. DOI: 10.19475 / j.carol carroll nki issn1674-957 - x. 2021.21.101.
[3] Li Heng. Intelligent vehicle control system based on CAN network [D]. Jilin University, 2020. DOI: 10.27162 /, dc nki. Gjlin. 2020. 006843.
[4] Chen Shihuan. Implementation Method of CAN Bus Communication in Design and Development of Vehicle Control System [J]. Modern manufacturing Technology and Equipment, 2022, 58 (01): 49-51. DOI: 10.16107 / j.carol carroll nki mmte. 2022. 0014.
[5] Chang Zhi. Research and Application of Vehicle Communication Protocol Based on CAN Bus [D]. North University of China,2019.
[6] WANG Rong. Intelligent Vehicle Data Acquisition and Processing based on CAN Bus [D]. Chongqing Jiaotong University, 2016.
[7] Fang Youyong. Multiple ECU based on CAN bus communication system design [j]. Journal of Nanchang University, 2020. DOI: 10.27232 /, dc nki. Gnchu. 2020. 003581.